

Milestone Accomplishment Record

River Hawk Consulting, LLC · IMAX HCD Prototype OT

August 8, 2026

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Milestone Accomplishment Record

Agreement No. 5600002690012 · River Hawk Consulting, LLC · UNCLASSIFIED Period of performance: **6 – 20 August 2026** (Article II)

Submitted under **Article VIII(b)**: *"Attachment 2 provides the mutually agreed-to acceptance criteria for each milestone. The Performer shall document for acceptance by the Government, the accomplishments of each milestone and shall submit such documentation to the AOR and AO for approval."*

Attachment 2 was not included in the executed agreement. The body references it four times (Article VIII(b) twice, Article XII once) but the LIST OF ATTACHMENTS names only Attachment 1. In its absence, the milestones and success indicators scored below are taken from the **TDD's own Milestones / Success Indicators table**, which is Attachment 1 and therefore incorporated. River Hawk will re-score against Attachment 2 on receipt if it differs. Attachment 3 — the reporting template Article XV(n) and XV(o) both name — is likewise referenced but absent.

Calendar

TDD Assumption (1): *"Day 1 begins the first calendar day after agreement execution."* The agreement was executed **6 August 2026**, so Day 1 is **7 August** and Day 14 is **20 August**, consistent with Article II.

Milestone	Day	Date	Status
M1 — proto-personas, empathy maps, task flows	2	Sat 8 Aug	✓ complete
M2 — repo scaffolded; five mock services containerized and smoke-deployed	4	Mon 10 Aug	✓ complete — run in Platforma
M3 — full wireframe set validated and frozen as build baseline	7	Thu 13 Aug	✓ complete
M4 — feature-complete UI wired to mock services; smoke suite passing	11	Mon 17 Aug	✓ complete
M5 — all six data deliverables transmitted; live demonstration provided	14	Thu 20 Aug	► this transmittal

River Hawk developed the demonstration software **at its own expense before award** (TDD, IP Assertion). The period of performance therefore opened with the build already complete, and M1, M3 and M4 were satisfied by work predating their nominal dates. This is stated plainly rather than presented as early delivery: the Government is receiving the benefit of pre-award investment, which is what the TDD said it would.

M1 — Proto-personas & empathy maps · Day 2

Success indicator (TDD): *"Two proto-personas with empathy maps and task flows delivered for Sponsor review."* **Verification (TDD)**: *"Artifacts in repo (optional Touchpoint 1 offered ~Day 3)."*

Evidence	Where
Four personas — two required, two secondary	hcd/personas.md
Empathy maps (Says / Thinks / Does / Feels)	hcd/empathy_maps.md
Four task flows and the end-to-end journey map	hcd/journey_map.md
20 per-persona success criteria, each naming the specification that replays it	hcd/journey_map.md

Delivered as `project/delivery/hcd_artifacts.pdf`. Exceeds the indicator: two personas were required, four are delivered.

Touchpoint 1 was offered and remains offered; the Sponsor has not scheduled it.

M2 — Repo, containerized services, sandbox smoke deployment · Day 4

Success indicator (TDD): *"GitHub repo scaffolded; all five mock services containerized and smoke-deployed to Platforma (or matched local K8s if sandbox access is pending)."* **Verification (TDD)**: *"Deployment log; running pods; repo access."*

The TDD wrote this verification assuming direct cluster access. The target environment is driven by **ArgoCD**, and River Hawk has no `kubectℓ` against it — deployment is a declarative sync, not a command someone runs. The equivalent evidence, and what should be captured instead, is:

TDD wording	ArgoCD equivalent
Deployment log	The Application's sync history — revision, sync result, timestamp
Running pods	The Application's health status and resource tree, showing all six workloads <code>Healthy</code>
Repo access	The Application's source repository and target revision

A single `argocd app get imax -o yaml` captures all three, or a screenshot of the Application view if only the UI is reachable. This is the same evidence in the form the environment actually produces it, and the Helm chart at `deploy/chart/` is what ArgoCD consumes.

Evidence	Where
GitHub repository, CI on every push	<code>Blackfell-Group/imax-chat-viewer</code>
Five schema-pinned mock services	<code>routes/</code> , <code>deploy/k8s/mock-*.yaml</code>
Dockerfiles and Kubernetes manifests	<code>deploy/Dockerfile.{spa, mock}</code> , <code>deploy/k8s/</code>
Helm chart for ArgoCD, proven equivalent to the manifests	<code>deploy/chart/</code> , <code>scripts/compare-chart-kustomize.py</code>
Six-image set built, scanned (trivy CRITICAL gate), deployed to a cluster and smoked — every push	<code>.github/workflows/angular-ci.yml</code>
Pod sizing within the TDD's 0.5 vCPU / 512 MB ceiling	250m CPU / 256Mi memory on all six

Status . The prototype has been **deployed and run in Platforma**, ahead of this milestone's Day 4 date. The ArgoCD Application view capture evidencing the sync and the running workloads is attached to the **high-side repository**, which is where it is producible: it is generated inside the enclave and River Hawk has no path to carry it low-side into this package. The Government verifies M2 against that capture.

Independently of the environment, the TDD's stated fallback — *"matched local K8s if sandbox access is pending"* — is also satisfied continuously and mechanically: CI re-proves the full chain on every push (build → trivy CRITICAL gate → deploy → rollout → in-cluster smoke). So M2 is evidenced twice, by different means.

M3 — Wireframes validated and frozen · Day 7

Success indicator (TDD): *"Full wireframe set validated with Sponsor input and frozen as build baseline."* **Verification (TDD):** *"Baselined wireframes; internal approval record (optional Touchpoint 2)."*

Evidence	Where
Persona-indexed annotated wireframes, every prototype screen	<code>hcd/wireframes.md</code>
Pain-to-design traceability: 24 pains → response → screen → specification → screenshot	<code>hcd/traceability_matrix.md</code>
Assumption log, 16 entries, 6 closed	<code>hcd/assumption_log.md</code>
Three workflow-model findings with rationale and restoration paths	<code>hcd/linguist_workflow_model.md</code> , <code>hcd/bilingual_display_model.md</code> , <code>hcd/one_output_model.md</code>

Sponsor input was not available. Wireframes were validated against the documented assumption log — the fallback the TDD states for exactly this case — and every assumption is flagged for Sponsor review in the acceptance package. Touchpoint 2 remains offered.

M4 — Feature-complete UI, green smoke suite · Day 11

Success indicator (TDD): *"Feature-complete UI wired to mock services; smoke suite passing on the full mission loop."* **Verification (TDD):** *"Green Playwright run; build demo (optional Touchpoint 3)."*

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Evidence	Where
Every wireframed screen implemented	<code>angular/src/app/features/</code>
Full mission loop: queue → search → thread → bilingual review → enrichment → OCR → promote to gold → export	<code>tests/ng/</code>
72 e2e specs green , plus 10 Angular unit and 93 Node unit specs	<code>npm run test:ng</code> , <code>npm test --prefix angular</code> , <code>npm run test:node</code>
Ported Phase 2 acceptance flows: 7 of 7 covered — 3 verbatim, 4 amended by logged HCD decisions	<code>project/evidence/README.md</code>
Machine-readable results	<code>project/evidence/e2e-results-*.json</code>

TDD Task 5 performance target verified. The TDD commits QA to "*<200 ms interaction latency on large logs via virtualization.*"

`tests/ng/virtual-scroll.spec.js` synthesizes a 2,000-message thread, asserts the DOM holds a bounded window, and measures worst scroll-to-repaint at **~17 ms**. This was previously unverified and the virtualization itself was previously absent; both were found and closed by the 7 August audit.

Twelve defects from internal review were closed on 7 August, each with a specification: resizable panels; a resizable, maximizable and zoomable scan viewer; session-wide document tags with an explicit Add control; right-to-left scan rendering; chronological message ordering with an ASC/DESC toggle; copy confirmation with a working manual fallback; a flagged-for-targeter facet and sort; a larger resizable export dialog; drag-reorderable gold copy that the export follows; and a retractable review verdict. Logged as QA in the board export and burndown — **not** claimed as Sponsor validation.

M5 — Acceptance package and live demonstration · Day 14

Success indicator (TDD): "*All six data deliverables transmitted; live demonstration provided.*" **Verification (TDD):** "*Acceptance package receipt.*"

Thursday 20 August 2026. All six data deliverables are transmitted with this package; the live demonstration is provided in person on the same date. Transmittal and demonstration are both required and neither substitutes for the other, so this milestone closes on the Government's receipt of the package and attendance at the demonstration — recorded here rather than asserted by the Performer.

The narrated walkthrough is delivered alongside the demonstration as the standing record of the working product, and as the fallback should enclave access be unavailable on the day.

Data deliverables — TDD D1–D6 crosswalk

The TDD names six data deliverables with formats and due days. This is the mapping the Government should use for acceptance.

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ID	Deliverable	Format required	Due	Artifact
D1	HCD Artifacts — personas, empathy maps, task flows/journey map, full wireframe set, assumption log, pain-to-design traceability matrix	PDF + source	D2 / D7 / D14	hcd/ (source) · project/delivery/hcd_artifacts.pdf
D2	UI Codebase — SPA implementing all HCD screens; Dockerfiles and Kubernetes charts for SPA and mock services; GitHub repository	Git repo	D4, then continuous	angular/, deploy/ on main; Blackfell-Group/imax-chat-viewer
D3	Test Suite — Playwright end-to-end smoke tests, test scripts, and final results report	In repo + PDF	D11 / D14	tests/ (source) · project/acceptance/test_results.md · project/delivery/test_results.pdf
D4	Documentation — user guide and developer guide, including production cutover path	PDF	D14	docs/ · project/delivery/{user_guide,developer_guide,deployment_guide,architecture}.pdf
D5	Project Artifacts — sprint-board export, burndown chart, risk register (final state)	PDF / CSV	D14	project/delivery/project_artifacts.pdf · project/delivery/board_export_final.csv
D6	Demo & Acceptance Package — recorded walkthrough (Sponsor receives unlimited rights),	MP4 / PDF + live	D14	project/acceptance/RiverHawk_IMAX_Walkthrough_2026-08-08.mp4 (4:51, narrated) · project/delivery/acceptance_deck.pdf · live demonstration 20 Aug

ID	Deliverable	Format required	Due	Artifact
	summary slide deck (objectives, test results, open issues), live demonstration			

Beyond D4. Article XII requires *"the architecture the Performer will deliver prior to the end of the period of performance"*, against the Article I(b) definition — sufficient *"to enable the Government to implement the system without requiring further development or support from the Performer."* Delivered as [docs/architecture.md](#) / [project/delivery/architecture.pdf](#). This is an Article XII obligation the TDD's D1–D6 list does not name; it was missed until the 7 August audit.

Administrative deliverables

Required by the agreement, outside the TDD's D1–D6.

Requirement	Article	Due	Status
FOCI package — SF 328, Key Management Personnel list, Organizational Entity Structure	XIV(c)	Not stated in the agreement	Identified; not submitted — disposition for the AO to direct
Cleared Personnel Certification Report (Attachment 3 template)	XV(n)	30 September annually	Identified; falls due after this period of performance — nil return expected
Cleared Personnel Disposition Report (Attachment 3 template)	XV(o)	Within 60 days of completion (~19 October 2026)	Identified; falls due after this period of performance — nil return expected

Article XV(a) scopes the security article to *"the extent that this Agreement involves access to national security information."* All work under this agreement is unclassified, no personnel are cleared under it, and the reports at XV(n) and XV(o) both fall due after the period of performance ends. River Hawk therefore expects nil returns on all three and has not submitted them with this package.

That expectation is stated rather than acted on. A nil return is still a return, and which of these the Agreements Officer wants — and in what form — is the Government's call, not the Performer's assumption. River Hawk will submit whatever the AO directs, on the schedule the AO sets.

Attachment 3, which XV(n) and XV(o) both name as the reporting template, is referenced by the agreement but was not included with it; the template is needed before either report can be submitted in the required form.